How the Kungsgatan and Slottsgatan Vissim Model Works

A plain-language guide to the model, its inputs, its behaviour and its results

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Model file	`Vissim_lab.inpx` (PTV Vissim)
Signal files	`Fixed_time.sig`, `Control_logic.vap`, `Signal_data.pua`
Field data	`traffic_measurements.xlsx` (10 weekday mornings)
Location	Kungsgatan and Slottsgatan, Norrköping

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About this document

This document explains, step by step, how the Vissim model of the signalised intersection of Kungsgatan and Slottsgatan in Norrköping works. It is written for readers who have never used Vissim: every technical term is explained when it first appears, and a glossary is given at the end (Section 11). Readers who know Vissim will find the exact object numbers and parameter values in the tables.

Every number in this document was read directly from the model files listed on the title page, not taken from memory or from general defaults. Where the document gives an estimate rather than a value from the files, it says so explicitly.

The model at a glance

What it represents	One four-armed signalised intersection in central Norrköping: Kungsgatan (north to south) crossing Slottsgatan (east to west).
Road users	Cars (95 %) and heavy goods vehicles (5 %), bicycles and e-bikes on two cycle tracks along Kungsgatan, pedestrians on four crosswalks.
Demand	2 150 motor vehicles, 200 cyclists and 170 pedestrians per hour, constant over time.
Signal control	Fixed-time, two stages, 90-second cycle (an actuated alternative is included).
How it works	Every road user is simulated individually, ten times per simulated second.
What it measures	Speeds and accelerations at 12 points, queue lengths at 10 points, and the average delay of all vehicles.
Simulated time	30 minutes (1 800 s) per run in the original file.

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1 What is a microscopic traffic simulation?

A *microscopic* traffic simulation represents every single road user (every car, truck, bicycle and pedestrian) as an individual object with its own position, speed, destination and personal driving style. This is different from *macroscopic* models, which describe traffic as a flow, like water in a pipe, without individual vehicles.

PTV Vissim, the software used here, moves time forward in small, equal steps. In this model the *simulation resolution* is 10 time steps per simulated second, so the simulation advances in steps of 0.1 s. In every step, each road user looks at its surroundings and decides how to move (Figure 1):

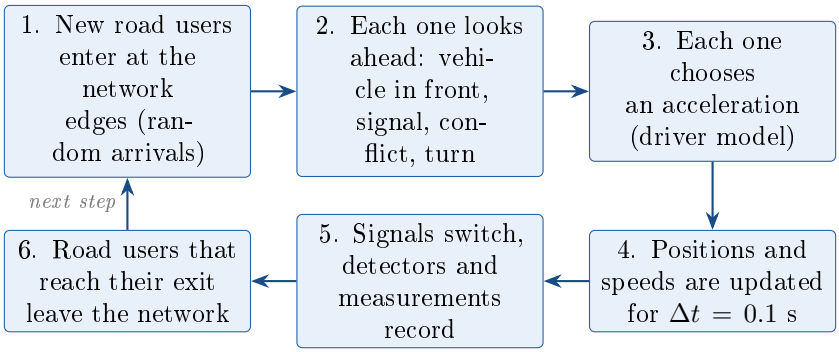


Figure 1. What happens in every simulation step of 0.1 s. One simulated half hour therefore consists of 18 000 such steps.

Randomness. Real traffic is never the same twice: drivers arrive at random moments and have different preferences. Vissim reproduces this with random numbers. Arrival times, each driver's preferred speed and each driver's personal following distance are drawn at random from statistical distributions. The sequence of random numbers is controlled by a *random seed*: the same model with the same seed always gives exactly the same result, while a different seed gives a different, equally plausible half hour of traffic. One simulation with one seed is called a *run* or *replication*. Because a single run is only one possible outcome, results are always averaged over several runs with different seeds (Section 8).

2 The road network

2.1 Links and connectors

In Vissim, roads are drawn as *links*: directed strips with one or more lanes on which road users travel in one direction only. Links are joined by *connectors*, short curved pieces that carry road users from the end of one link onto another link. All turning movements inside the intersection are connectors. The model contains 22 links and 16 connectors (Table 1). The network is drawn for right-hand traffic.

Every motorised approach has a single lane of 3 m. A short *left-turn pocket* branches off each approach some 10 to 20 m after the network edge, so that vehicles waiting to turn left do not block vehicles going straight or turning right. Kungsgatan is modelled as two continuous links (2 and 4) that run straight through the junction; Slottsgatan ends at the junction on each side and continues on separate exit links (1 and 3). Figure 2 shows the network as drawn from the coordinates in the file, together with the measurement points described in Section 7.

Table 1. The 22 links of the model. "Behaviour" is the link behaviour type, which decides how road users on that link interact (Section 4). Lengths are measured from the link geometry in the file.

Link	Role	Behaviour	Width	Length
2	Kungsgatan northbound, from the south edge through the junction to the north edge	Urban	3.0 m	131 m
4	Kungsgatan southbound, from the north edge through the junction to the south edge	Urban	3.0 m	115 m
5	Slottsgatan westbound approach (traffic arriving from the east)	Urban	3.0 m	59 m
11	Slottsgatan eastbound approach (traffic arriving from the west)	Urban	3.0 m	63 m
1	Slottsgatan exit towards the east	Urban	3.0 m	68 m
3	Slottsgatan exit towards the west	Urban	3.0 m	63 m
6, 8, 7, 12	Left-turn pockets of the northbound, southbound, westbound and eastbound approaches	Urban	3.0 m	30 to 37 m
9, 10, 13 to 18	Pedestrian crosswalks, two directions on each of the four arms	Footpath	1.5 m	14 to 15 m
19 to 22	Cycle tracks along Kungsgatan, two per direction (before and after the junction)	Cycle track	1.5 m	58 to 63 m

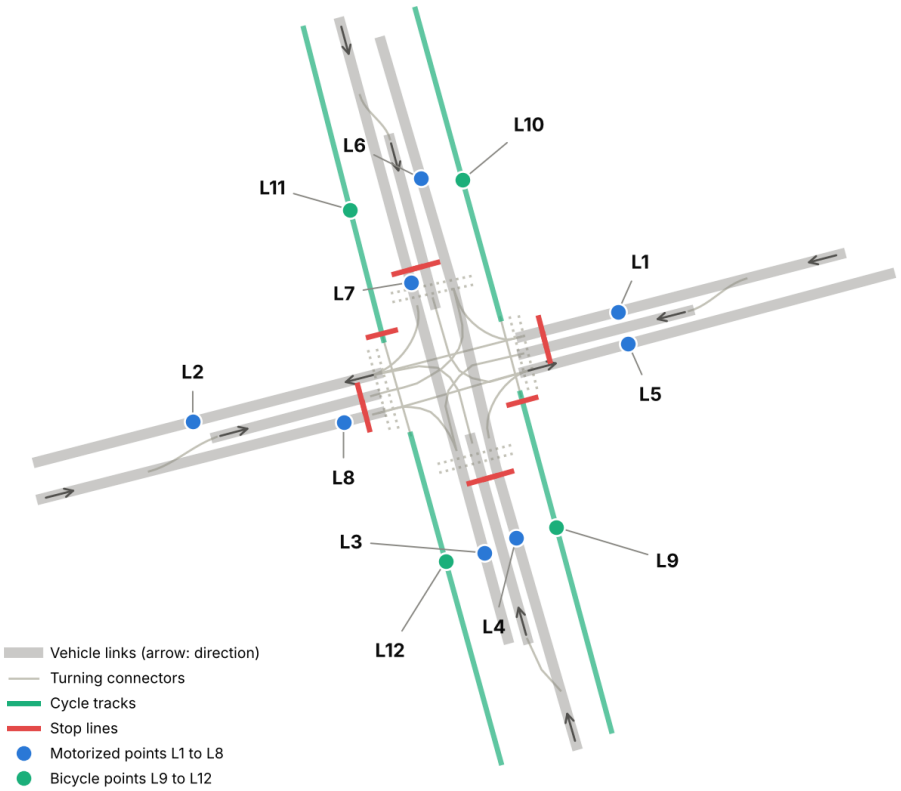


Figure 2. The Vissim network: links, turning connectors, stop lines and the twelve data collection points L1 to L12, plotted from the coordinates in `Vissim_lab.inpx`.

2.2 Turning movements

The 16 connectors create every possible movement through the junction. Right turns and the straight movements on Slottsgatan go directly from the approach to the exit; left turns go from the approach into the left-turn pocket and from there across the junction (Table 2). The two cycle tracks only go straight.

Table 2. Movements and the connectors that carry them (connector numbers as in the model file).

Approach	Left	Straight	Right
Kungsgatan northbound (2)	10003, pocket 6, 10006 to west	link 2 continues	10000 to east
Kungsgatan southbound (4)	10010, pocket 8, 10009 to east	link 4 continues	10007 to west
Slottsgatan westbound (5)	10002, pocket 7, 10005 to south	10001 to west	10004 to north
Slottsgatan eastbound (11)	10013, pocket 12, 10012 to north	10011 to east	10008 to south
Cycle tracks (20, 22)		10014, 10015	

3 Who uses the network: traffic demand

3.1 Types of road users

Vissim groups road users into *vehicle types*, which are collected in *vehicle classes*. Table 3 lists the types that actually appear in this model. Pedestrians are modelled in a simple way, as a special vehicle type that walks on the crosswalk links.

Table 3. Road user types that appear in the model.

Class	Type (number)	Where	Share	Desired speed
Car (10)	Car (100)	all motorised inputs	95 %	uniform 48 to 58 km/h
HGV (20)	HGV (200)	all motorised inputs	5 %	uniform 48 to 58 km/h
Bike (60)	Bike (610)	both cycle tracks	80 %	uniform 12 to 15 km/h
Bike (60)	E-bike (620)	both cycle tracks	20 %	uniform 15 to 20 km/h
Pedestrian (50)	Man (510)	all crosswalks	100 %	uniform 4 to 6 km/h

3.2 How road users enter the model

Road users are created at *vehicle inputs*, placed at the start of each link that enters the network. An input has a volume in road users per hour and a *vehicle composition*, which says which types arrive and in what proportions. The arrivals are *stochastic*: the input produces the stated number per hour on average, but the exact moments of arrival are random, as in reality. The model has 14 inputs (Table 4). The volumes are defined for three consecutive 15-minute intervals with identical values, so the demand is constant over the simulated period.

3.3 Where they go: routes

Once a vehicle has entered, a *static routing decision* near the start of the approach assigns it a route, i.e. left, straight or right. The choice is random, with fixed probabilities given by the relative flows of the routes (Table 5). Multiplying the input volume by these shares gives the average hourly volume of each movement.

Kungsgatan carries four fifths of the motor traffic. The heaviest single movement is the 780 vehicles per hour going straight north on Kungsgatan. On Slottsgatan, most vehicles turn; only 8 to 14 % go straight across.

Table 4. Vehicle inputs of the model. Letters A, C, E and G are the names used in the lab instructions.

Input	Location	Link	Per hour	Composition
1	A: Kungsgatan from the north (southbound)	4	660	Default (95 % car, 5 % HGV)
3	E: Kungsgatan from the south (northbound)	2	1 040	Default
4	C: Slottsgatan from the east (westbound)	5	240	Default
11	G: Slottsgatan from the west (eastbound)	11	210	Default
14, 15	Cycle tracks, north- and southbound	20, 22	100 each	Bike (80 % bike, 20 % e-bike)
5, 6	Crosswalk on the south arm, both directions	13, 14	25 each	Ped
7 to 10, 12, 13	Crosswalks on the other three arms	9, 10, 15 to 18	20 each	Ped

Table 5. Route shares of the four routing decisions and the resulting average movement volumes (vehicles per hour).

		Share			Volume [veh/h]		
Approach	**Input**	**Left**	**Straight**	**Right**	**Left**	**Straight**	**Right**
Kungsgatan northbound (E)	1 040	0.08	0.75	0.17	83	780	177
Kungsgatan southbound (A)	660	0.15	0.79	0.06	99	521	40
Slottsgatan westbound (C)	240	0.50	0.08	0.42	120	19	101
Slottsgatan eastbound (G)	210	0.57	0.14	0.29	120	29	61

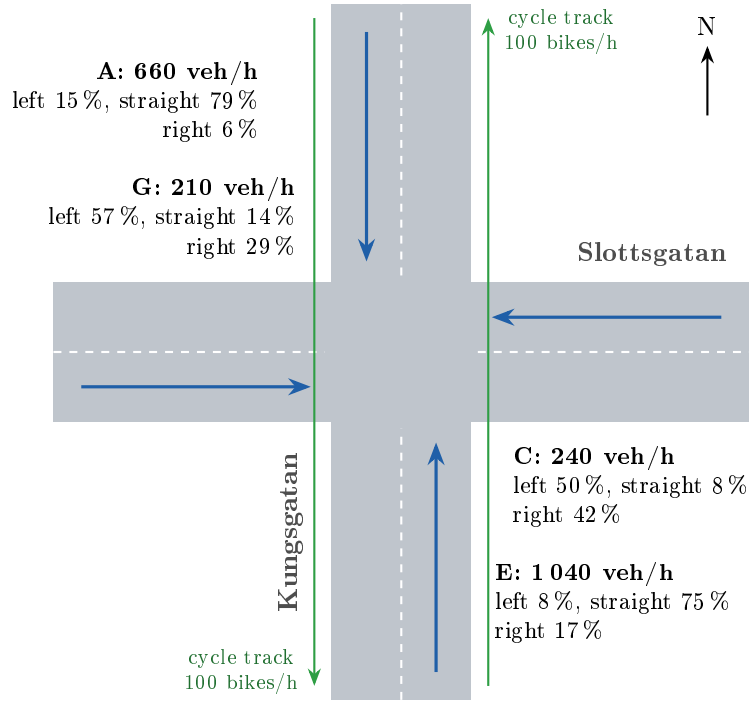


Figure 3. Schematic of the demand (not to scale). Blue arrows: motorised approaches with their hourly volume and turning shares. Green arrows: the two cycle tracks along Kungsgatan.

4 How road users move

This section describes the rules that decide how each road user accelerates, follows others, brakes and turns. Together they are called the *driving behaviour*.

4.1 Desired speed

Every road user has a *desired speed*: the speed it would choose on an empty road. It is drawn at random from a *desired speed distribution* when the road user enters the network and kept for the whole trip. In this model all distributions used for road users are uniform, meaning every value between the lower and the upper bound is equally likely (Table 3). A car therefore wants to drive somewhere between 48 and 58 km/h, on average 53 km/h. A road user drives slower than its desired speed only when something forces it to: a vehicle in front, a red signal, a turn or a conflict.

4.2 Acceleration and braking

How quickly a vehicle speeds up or slows down is given by speed-dependent functions. The *desired acceleration* is what a driver normally uses; the *maximum acceleration* is the technical limit of the vehicle. Both decrease with speed: a car accelerates strongly from standstill and more gently near its desired speed (Figure 4). Each vehicle also receives its own random position between a lower and an upper curve, so that some drivers are more dynamic than others.

For normal braking, cars use a *desired deceleration* of -2.75 m/s^2 and HGVs -1.25 m/s^2, independent of speed. In an emergency, cars can brake up to -7.5 m/s^2 and HGVs up to -5.5 m/s^2 (the *maximum deceleration*).

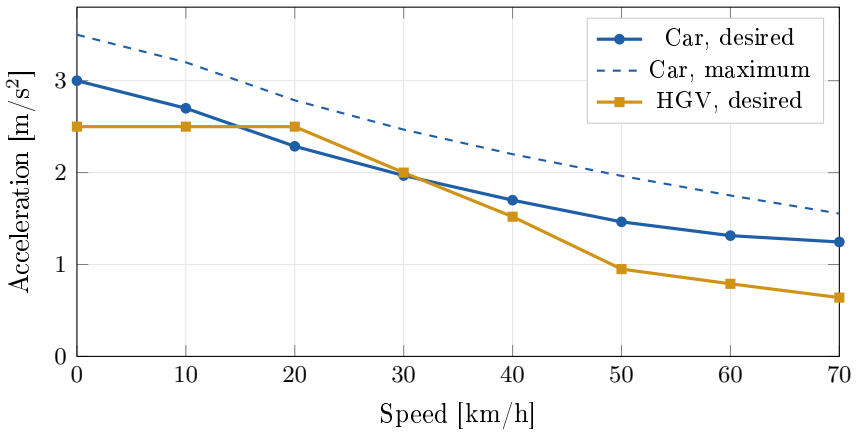


Figure 4. Average desired and maximum acceleration as a function of speed, read from the acceleration functions in the model file.

4.3 Following the vehicle in front: the Wiedemann 74 model

The most important rule is how a vehicle follows the vehicle in front of it. All motorised links use the driving behaviour *Urban (motorized)*, which applies the car-following model of Wiedemann (1974), the standard model for urban traffic in Vissim. Its basic idea is that a driver is always in one of four situations:

1. **Free driving:** nobody close in front; the driver accelerates towards the desired speed.
2. **Approaching:** a slower vehicle is ahead; the driver slows down to match its speed before reaching the safety distance.

3. **Following:** the driver follows at about the safety distance, with small, unconscious speed fluctuations around the leader's speed.
4. **Braking:** the gap has become shorter than the safety distance; the driver brakes more strongly to restore it.

The *safety distance* d that a driver wants to keep to the vehicle in front is

$$d = a_x + b_x, \qquad b_x = (b_{x,\text{add}} + b_{x,\text{mult}} \cdot z)\,\sqrt{v}, \tag{1}$$

where v is the speed in m/s and z is a personal random value of each driver between 0 and 1 (on average 0.5). The three parameters of Equation (1) are the ones that the lab instructions suggest for calibration (Table 6).

Table 6. Car-following parameters of driving behaviour 1, Urban (motorized), as given in the model file.

Parameter	Value	Meaning
a_x, average standstill distance	2.0 m	Gap between two stopped vehicles, e.g. in a queue at a red light.
$b_{x,\text{add}}$, additive part of safety distance	2.0	Base part of the extra distance that grows with speed.
$b_{x,\text{mult}}$, multiplicative part	3.0	Personal part of the extra distance, scaled by the random value z; it makes drivers different from each other.
Reduction factor near signals	0.6	Within 100 m upstream and 100 m downstream of a signal head, the safety distance is multiplied by 0.6, because drivers close up when approaching and leaving a stop line.
Look-ahead distance	0 to 250 m	How far ahead a driver perceives vehicles and signals; up to 4 objects ahead are taken into account.

Worked example

A car driving at 50 km/h ($v = 13.9$ m/s, $\sqrt{v} = 3.73$) with an average personal value $z = 0.5$ wants a safety distance of

$$d = 2.0 + (2.0 + 3.0 \cdot 0.5) \cdot 3.73 = 2.0 + 13.0 = 15.0 \text{ m},$$

which corresponds to a time gap of about 1.1 s. Close to the signal the extra part is reduced to 60 %, giving $2.0 + 0.6 \cdot 13.0 \approx 9.8$ m. Smaller safety distances let more vehicles pass the stop line per second of green, so these parameters directly control the *capacity* of the intersection.

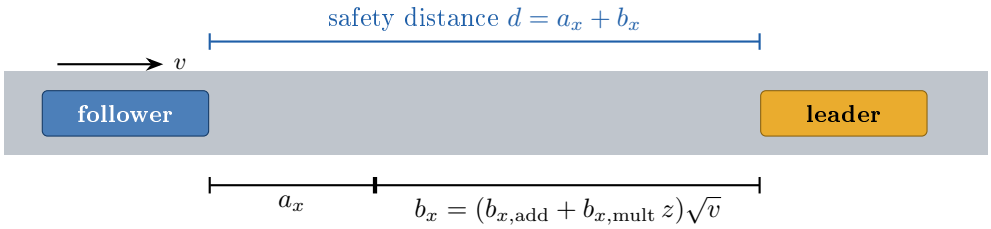


Figure 5. The safety distance in the Wiedemann 74 car-following model: a fixed standstill part a_x and a part b_x that grows with the square root of the speed v.

4.4 Cyclists and pedestrians

Cyclists use the behaviour *Cycle-Track (free overtaking)*, which applies the Wiedemann 99 model with a standstill distance of 0.5 m and allows cyclists to overtake each other within the 1.5 m wide track. Pedestrians use *Footpath (no interaction)*: they walk at their desired speed without reacting to each other. Both groups do, however, stop at red signals and have priority over turning vehicles (Section 5).

4.5 Slowing down for turns: reduced speed areas

Vehicles cannot take a sharp turn at 50 km/h. Vissim handles this with *reduced speed areas*: short sections on the turning connectors where each vehicle class receives a lower desired speed and approaches it with a given deceleration. Outside the area, the vehicle returns to its own desired speed. Table 7 lists the eight areas of the model.

Table 7. Reduced speed areas on the turning connectors (speeds in km/h, uniform distributions; deceleration towards the area 2 m/s^2).

Area	Connector	Movement	Cars	HGV
1	10000	right turn from Kungsgatan northbound	20 to 25	25 to 30
5	10007	right turn from Kungsgatan southbound	20 to 25	25 to 30
4	10004	right turn from Slottsgatan westbound	20 to 25	25 to 30
9	10008	right turn from Slottsgatan eastbound	25 to 30	20 to 25
10	10012	left turn from Slottsgatan eastbound	25 to 30	20 to 25
6, 7, 8	10005, 10006, 10009	other three left turns	none	none

Two details of the original file are worth noting. First, in areas 1, 4 and 5 HGVs are allowed to turn *faster* than cars, while areas 9 and 10 have the opposite order. Second, areas 6, 7 and 8 contain no speed at all, so left-turning vehicles on those connectors are slowed only by the curve geometry and by the vehicles they must give way to. Because the areas lie on the connectors, that is after the stop line, a turning vehicle starts to slow down only shortly before the turn.

5 Who goes first: conflict areas

Wherever two paths cross or merge inside the intersection, Vissim places a *conflict area*. A conflict area tells the road users on the two paths who has priority. The road user without priority checks whether the conflict area is free for long enough (a *gap*) and waits if it is not. The model contains 91 conflict areas (Table 8).

Table 8. The 91 conflict areas of the model, grouped by the road users involved.

Paths involved	Number	Rule
Vehicle and pedestrian	37	In 33 the pedestrian has priority; 4 are passive.
Vehicle and cyclist	12	The cyclist always has priority.
Vehicle and vehicle, with priority rule	8	Left-turning vehicles give way to oncoming vehicles going straight or turning right (a *permitted* left turn).
Vehicle and vehicle, passive	34	No priority rule: the two streams never have green at the same time, so the signals already separate them.
Total	91	All with front and rear gap 0.5 s and safety distance factor 1.5.

In practice this means: a car turning right from Kungsgatan must wait for cyclists going straight on the cycle track next to it, and for pedestrians on the crosswalk it crosses after the turn. A car turning left must in addition wait for a gap in the oncoming traffic. This matches the lab description, in which the conflict between turning vehicles and pedestrians is resolved by giving preference to pedestrians.

6 Traffic signals

6.1 Signal groups and stages

A *signal group* is a set of signal heads that always show the same colour. The controller has five signal groups (Table 9). Signal heads are placed at the stop line of every lane and crosswalk, 18 in total.

Table 9. Signal groups, the road users they control and the stage in which they show green.

SG	Name	Controls	Colour sequence	Green in
1	Kungsgatan	vehicles on Kungsgatan (links 2, 4, 6, 8)	red, red/amber, green, amber	stage 1
5	Kungsgatan_bike	cyclists on both cycle tracks	red, green	stage 1
4	Slottsgatan_ped	pedestrians crossing Slottsgatan (9, 10, 17, 18)	red, green	stage 1
3	Slottsgatan	vehicles on Slottsgatan (links 5, 7, 11, 12)	red, red/amber, green, amber	stage 2
2	Kungsgatan_ped	pedestrians crossing Kungsgatan (13 to 16)	red, green	stage 2

The groups are combined into two *stages*. In stage 1, Kungsgatan has green: its vehicles, the cyclists riding along it and the pedestrians walking along it (who cross Slottsgatan). In stage 2, Slottsgatan has green, together with the pedestrians crossing Kungsgatan. Between the stages there is an *interstage*, a short transition in which the ending groups turn amber and red and the starting groups wait until the conflict zone has been cleared. The minimum clearing times are stored in the *intergreen matrix*: 5 s after a vehicle group ends and 7 s after a pedestrian or bicycle group ends, before a conflicting group may start.

6.2 The fixed-time program

In the calibration and baseline situation the controller runs signal program 1 of `Fixed_time.sig` ("Kungsgatan, Stage 1, Slottsgatan, Stage 2"). Every cycle lasts 90 s and is exactly the same. The interstage from stage 1 to stage 2 starts at second 33 and lasts 11 s; the interstage back starts at second 73 and lasts 14 s. Reading the individual signal commands gives the timing in Figure 6.

Table 10. Green times per 90 s cycle, read from the signal program.

Signal group	Green	Amber	Share of cycle
Kungsgatan vehicles (SG 1)	45 s	3 s	53 %
Slottsgatan vehicles (SG 3)	35 s	3 s	42 %
Bicycles, pedestrians across Slottsgatan (SG 5, 4)	43 s		48 %
Pedestrians across Kungsgatan (SG 2)	33 s		37 %

The signal file also contains a second program ("Stage 1, Stage 2, Opt") that shortens the Slottsgatan stage by 10 s; it is not used in the model.

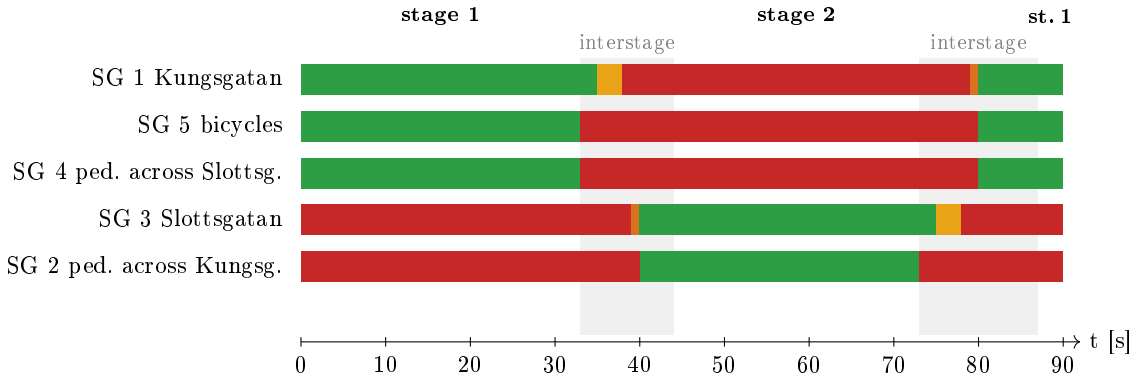


Figure 6. Fixed-time signal plan (program 1, cycle 90 s) as read from `Fixed_time.sig`. Green, amber and red bars show the signal colour; grey shading marks the two interstages.

How busy is the busiest approach? (estimate)

A single urban lane can discharge roughly 1 800 to 1 900 vehicles per hour of green (a typical *saturation flow*; this value is an assumption, not a number from the file). Kungsgatan northbound has 45 s of green plus 3 s of amber out of 90 s, so its capacity is roughly

$$1\,900 \cdot \frac{45 \text{ to } 48}{90} \approx 950 \text{ to } 1\,010 \text{ vehicles per hour.}$$

The vehicles that stay in the main lane (straight and right, $780+177 = 957$ per hour) therefore use 95 to 100 % of the capacity. This approach is *near saturation*: small changes in driver behaviour or demand can create long queues. All other approaches have plenty of spare capacity.

6.3 The actuated alternative

The lab also provides an *actuated* controller, written in the VisVAP language (`Control_logic.vap`) together with the interstage definitions in `Signal_data.pua`. It uses the same two stages and the same interstages, but the length of each stage depends on the traffic detected. Eight detectors, each 3 m long, are placed on the approaches and in the pockets and send their signals to six input channels (Table 11).

Table 11. Detectors used by the actuated controller.

Det.	Link	Location	Channel	Distance to stop line
1	2	Kungsgatan northbound, main lane	4	26 m
2	6	Kungsgatan northbound, left pocket	5	4 m
4	4	Kungsgatan southbound, main lane	1	26 m
3	8	Kungsgatan southbound, left pocket	2	4 m
6	5	Slottsgatan westbound, main lane	3	4 m
5	7	Slottsgatan westbound, left pocket	3	4 m
7	11	Slottsgatan eastbound, main lane	6	4 m
8	12	Slottsgatan eastbound, left pocket	6	4 m

The logic is evaluated once per second and follows the decision chart in Figure 7. Stage 1 (Kungsgatan) is *extended* as long as vehicles keep arriving on any Kungsgatan channel with a time gap of at most 3 s, up to 75 s. When the gaps become longer, or the maximum is reached, the controller switches to Slottsgatan only if a vehicle is waiting there (a *call* on channel 3 or 6);

otherwise Kungsgatan simply keeps green. Stage 2 (Slottsgatan) is extended in the same way but for at most 15 s, and then the controller returns to Kungsgatan.

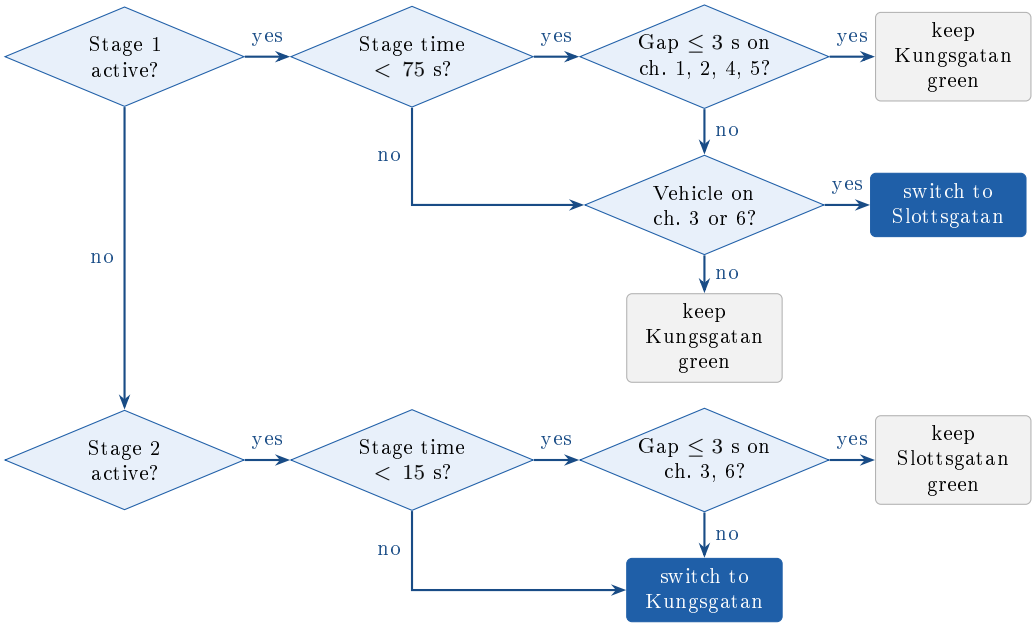


Figure 7. Logic of the actuated controller, simplified from `Control_logic.vap`. The check runs once per second; during an interstage nothing is decided.

7 What the model measures

The model contains three kinds of measurement objects, which record what happens during a run.

7.1 Data collection points: speeds and accelerations

A *data collection point* works like a loop detector in the road: it records every vehicle that passes and its speed and acceleration at that moment. The model has twelve, called L1 to L12, at the same places where the field data were collected (Table 12). The position relative to the stop line decides what a point actually sees: points shortly before a stop line see vehicles braking or queueing, points after the junction see vehicles accelerating away.

Because left-turners leave the main lane into the pocket before L1, L4, L7 and L8, these points see only straight and right-turning vehicles. At L1 and L8 most of them are right-turners ($101/120 = 84\,\%$ and $61/90 = 67\,\%$), so the right-turn speed strongly influences these points.

7.2 Queue counters

A *queue counter* measures the length of the queue upstream of its position. A vehicle is counted as queueing when its speed falls below 5 km/h and stops being counted when it exceeds 10 km/h; vehicles more than 20 m apart are not considered part of the same queue, and queues are followed up to 500 m. The model has ten queue counters: one at each of the eight vehicle stop lines (main lane and left pocket of every approach) and one on each cycle track. They report the average queue length, the maximum queue length and the number of stops in the queue.

7.3 Vehicle network performance

This evaluation summarises the whole network in a few numbers, most importantly the *average delay per vehicle*. The delay of a trip is its actual travel time minus the travel time it would have

Table 12. The twelve data collection points (Vissim data collection measurements 1 to 12) and what they mainly capture.

Point	Name in the model	Link	Position	Mainly captures
L1	Slottsgatan_East_in	5	14.0 m before stop line	braking and queueing; straight and right only
L2	Slottsgatan_West_out	3	34 m along the exit	acceleration after the junction
L3	Kungsgatan_South_out	4	52.6 m after stop line	acceleration towards desired speed
L4	Kungsgatan_South_in	2	11.7 m before stop line	queue on the busiest approach
L5	Slottsgatan_East_out	1	19 m along the exit	acceleration after the junction
L6	Kungsgatan_North_out	2	55.3 m after stop line	acceleration towards desired speed
L7	Kungsgatan_North_in	4	2.1 m *after* stop line	discharge from the stop line
L8	Slottsgatan_West_in	11	4.1 m before stop line	braking and queueing; straight and right only
L9 to L12	Kungsgatan bike points	19 to 22	on the cycle tracks	cyclist speeds

had without other vehicles, signals or stops. Average delay is the usual measure of the *level of service* of a signalised intersection. The same evaluation also gives the number of stops per vehicle, the average speed and the number of vehicles that completed their trip.

8 Running the model

Table 13 lists the simulation settings stored in the original file.

Table 13. Simulation settings of the original model file.

Setting	Value	Meaning
Simulation period	1 800 s	30 minutes of traffic are simulated.
Simulation resolution	10	Time steps per simulated second (step of 0.1 s).
Number of runs	5	Replications performed one after another.
Random seed	42	Starting value of the random number generator.
Random seed increment	1	Each following run uses the next seed (42, 43, 44, ...).
Evaluation interval	0 to 1 800 s	Results are aggregated over the whole period.

Warm-up. At the start of a run the network is empty. During the first minutes there are fewer vehicles and shorter queues than in normal operation. In the original file the measurements start at second 0, so this unrealistic start-up phase is included in the results. In the calibration work (Section 9) the period was extended to 2 400 s and the first 600 s were excluded as *warm-up*.

How many runs? Because every run is one random outcome, the mean of several runs is used. The number of runs n needed for a mean $\bar{x}$ with relative accuracy ε follows the course formula

$$n = \left(\frac{t_{0.975,\, n_0 - 1}\; s}{\varepsilon\, \bar{x}} \right)^2 , \tag{2}$$

where s is the standard deviation between n_0 pilot runs and t the Student t value. For $\varepsilon = 5\,\%$ the points after the junction need only one or two runs, but the low-speed approach points L1 and L8 need about 10 to 14 runs, because few vehicles pass them and their speed depends on whether they arrive on red or green.

9 Is the model realistic? Field data and calibration

9.1 The field data

The file `traffic_measurements.xlsx` contains the mean speed at L1 to L12 and the mean acceleration at L1 to L8, measured on ten weekday mornings under the current fixed-time control and the current demand. A model is accepted at a point if a two-sample t-test (95 % confidence) cannot distinguish the simulated mean from the field mean. The measured variation between days is very different between points: the standard deviation of the mean speed is 4.7 km/h at L1 but only 0.3 km/h at L3.

9.2 How well the original model fits

With a 600 s warm-up and ten replications, the original model reproduces the accelerations well (7 of 8 points pass the test) but only 2 of the 12 speeds. The main differences have clear causes in the model mechanics described above:

- **L1 and L8 are too fast** (by 10.7 and 4.7 km/h): most vehicles there turn right, but the reduced speed areas lie after the stop line (Section 4.5), so they pass the measurement point almost at full speed.
- **L4 and L7 are too fast** (by about 5 km/h): approach speed and queue discharge on Kungsgatan are higher than observed.
- **Cyclists are too slow** (by about 3.8 km/h): the model's mean desired speed is $0.8 \cdot 13.5 + 0.2 \cdot 17.5 = 14.3$ km/h, while about 18 km/h is observed.

9.3 The calibrated model

The model was calibrated automatically: Python generated hundreds of model variants with different parameter values, ran them in Vissim, learned a statistical *surrogate model* (a Gaussian process) that predicts the results from the parameters, and used it to find the best parameter values, which were then checked with new random seeds. The recommended calibrated model changes the parameters in Table 14.

With these values the average speed error over the twelve points falls from 4.53 to 2.01 km/h, the acceleration error stays at the original level (0.13 m/s^2), all four cycle-track points and the points L6, L7 and L8 pass the test (10 of 20 tests in total), and the full demand is still served. Figure 8 compares the original and the calibrated model point by point.

9.4 What the calibrated model was used for

The calibrated model is the baseline for comparing designs under identical conditions (20 runs each with the same random seeds). Table 15 summarises the three alternatives of the lab.

10 Simplifications and limitations

The lab instructions state that the model is a simplification of reality made for teaching. A reader should keep the following points in mind:

Table 14. Original and calibrated parameter values (recommended model).

Parameter	Original	Calibrated	Interpretation
Bicycle / e-bike desired speed	13.5 / 17.5	17.3 / 21.3	cyclists ride faster than assumed
Right-turn speed (cars)	22.5	10.1	right turns are taken slowly
Left-turn speed (cars)	(27.5)	29.9	now defined on all left turns
Desired speed Slottsgatan from the east	53	43.9	a slower street than the others
Desired speed other approaches	53	51.3 to 52.4	close to the original
a_x / $b_{x,\text{add}}$ / $b_{x,\text{mult}}$	2.0 / 2.0 / 3.0	1.49 / 0.82 / 1.39	tighter following keeps the busy approach at capacity
Reduction factor near signals	0.6	0.31	drivers close up more near the stop line
Desired acceleration / deceleration scale	1.0 / 1.0	1.04 / 0.89	almost unchanged

Speeds in km/h, centre of a uniform distribution. Each motorised input received its own desired speed distribution.

Table 15. Results of the alternative designs compared with the calibrated baseline (average delay of all vehicles, 20 runs).

Scenario	Delay [s/veh]	What happens
Baseline (fixed time)	21.3	Reference.
Actuated control (Section 6.3)	15.7	26 % less delay: unused Slottsgatan green is given to the near-saturated Kungsgatan; Kungsgatan queues are halved.
Incident demand (volumes change every 10 min)	21.3	No change on average, but Slottsgatan left-turn queues double in minutes 10 to 20 and then dissolve.
800 bicycles per hour and direction	22.5	Right-turning cars wait for cyclists; car flow on Kungsgatan drops by 7 % and stops double.

- Each approach has one lane plus a left-turn pocket; there are no gradients, parking, buses or other intersections nearby.
- The demand is constant over the half hour and there are no field counts of traffic volume, so the demand is taken as given.
- Pedestrians are modelled as a simple vehicle type that walks without interacting with other pedestrians, and all of them are of the type “Man”.
- Turning speeds act only on the connectors, i.e. after the stop line, and three of the four left-turn areas have no speed in the original file (Section 4.5).
- The original file has no warm-up period, which biases results towards an emptier network.
- The model is calibrated for 100 cyclists per hour; results far outside this situation, such as 800 cyclists per hour, are less certain because the car and cyclist interaction could not be calibrated with the available data.

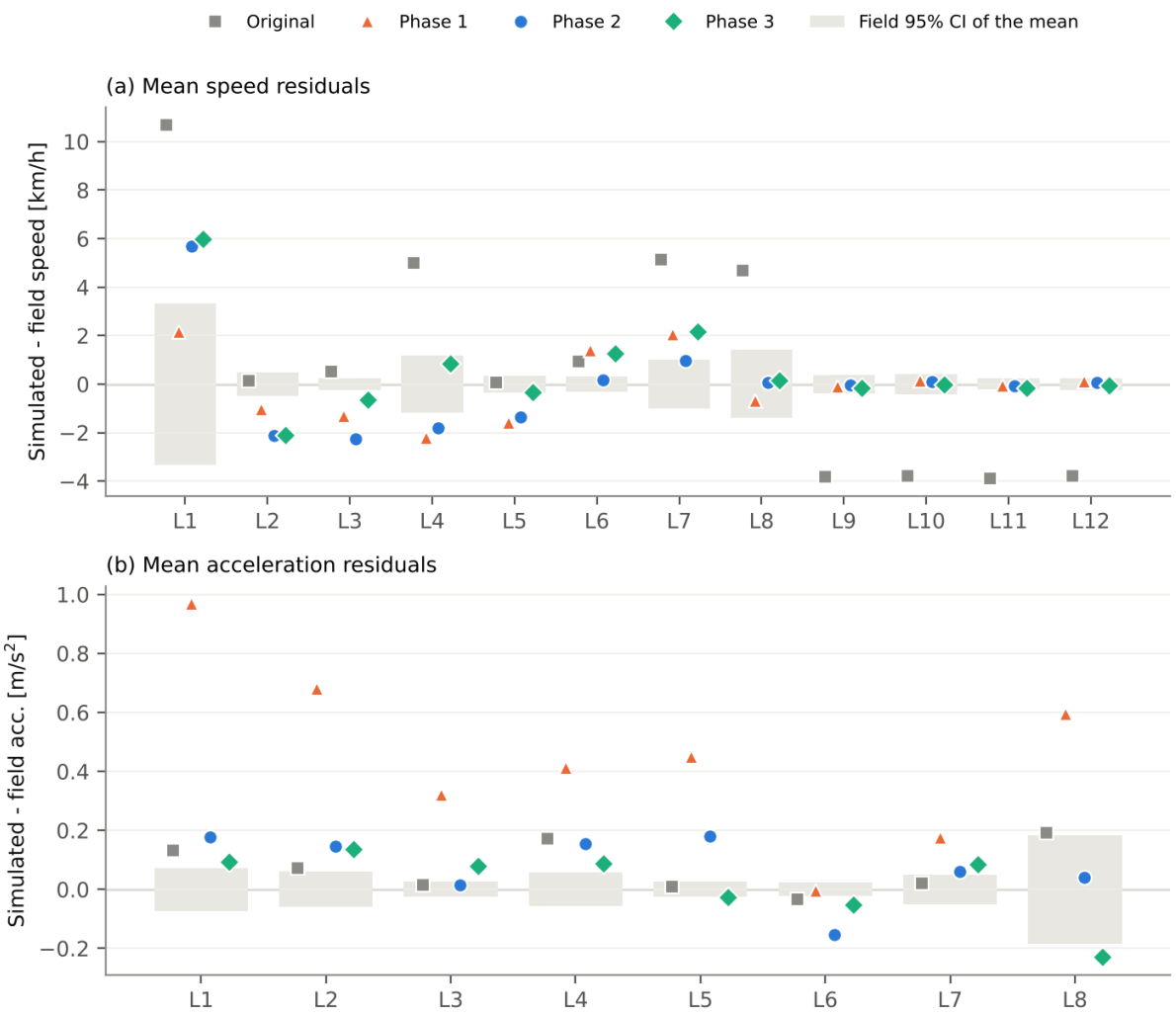


Figure 8. Simulated minus field mean for (a) speeds and (b) accelerations, ten independent runs. The grey bars show the 95 % confidence interval of the field mean. The phase 2 model is the recommended calibrated model.

11 Glossary

Calibration
Adjusting model parameters until simulated results agree with field measurements.

Capacity
The largest number of vehicles that can pass a point per hour.

Conflict area
An area where two paths cross or merge, with a rule for who goes first.

Connector
A short link piece that joins two links; all turns are connectors.

Cycle time
Time for one complete sequence of all signal stages (here 90 s).

Data collection point
A virtual detector that records passing vehicles, their speed and acceleration.

Delay
Actual travel time minus the travel time without other traffic, signals or stops.

Desired speed
The speed a road user would choose on an empty road.

Detector
A sensor in the road that tells an actuated controller that a vehicle is present.

Interstage
The transition between two signal stages, with amber and clearance time.

Link A directed road section with one or more lanes.

Queue counter
An object that measures the length of the queue upstream of it.

Random seed
Starting value for the random numbers; the same seed gives the same run.

Reduced speed area
A section where road users temporarily use a lower desired speed, e.g. in a turn.

Replication (run)
One simulation of the period with one random seed.

Saturation flow
The rate at which a queue discharges during green, in vehicles per hour of green.

Signal group
A set of signal heads that always show the same colour.

Stage
A combination of signal groups that have green at the same time.

Surrogate model
A fast statistical model that imitates the results of the slow simulation.

Vehicle composition
The mix of vehicle types arriving at an input.

Vehicle input
The point where road users enter the network, with a volume per hour.

Warm-up
The initial part of a run that is not evaluated, because the network starts empty.

Wiedemann 74 / 99
Car-following models in Vissim that describe how a driver follows the vehicle in front.

A Where to find each element in Vissim

Table 16. Model elements described in this document and where to find them in the Vissim user interface.

Element	Vissim menu or object
Links and connectors	Network objects sidebar: Links
Vehicle inputs	Network objects sidebar: Vehicle Inputs, show list
Routes	Network objects sidebar: Vehicle Routes, Static, show list
Vehicle compositions	Traffic: Vehicle Compositions
Desired speed distributions	Base Data: Distributions: Desired Speed
Acceleration functions	Base Data: Distributions (acceleration and deceleration functions)
Car-following parameters	Base Data: Driving Behaviors: Urban (motorized)
Reduced speed areas	Network objects sidebar: Reduced Speed Areas
Conflict areas	Network objects sidebar: Conflict Areas
Signal controller	Signal Control: Signal Controllers (VISSIG editor for `Fixed_time.sig`)
Detectors	Network objects sidebar: Detectors
Data collection points	Network objects sidebar: Data Collection Points; Evaluation: Measurement Definition: Data Collection Measurements
Queue counters	Network objects sidebar: Queue Counters
Simulation settings	Simulation: Parameters
Evaluation settings	Evaluation: Configuration: Result Attributes